Chapter 3
Identification Test Design

The necessity and possibility of test in control

- The purpose of identification test (for control) is to excite and to collect (control) relevant information.
- Test signals/excitations are used during the tests.
- Identification tests is a cost, but do not disturb normal production.
- A good test is the key to a successful identification/control.
- Without a good test identification is data-in-garbage-out.

3.2 Preliminary Tests

The purpose is to collect *a priori* knowledge of the process.

3.2.1 Collect historic data

- To get a first impression of the process.
- The outputs (CVs) of the process represent the unmeasured disturbances.
- The movements of MVs represent operators control.
- Easy and cheap. Databases are available for most of processes.
- Talk to and learn from operators.

Remark: In general historic data is not rich enough for model identification, or, put it in other way, the signal-to-noise ratio is too low (e.g. 90% noise, 10% signal)

3.2 Preliminary Tests

3.2.2 Short step test

- Step each MV up and down a few times.
- Check the dominant time constants, gains and delays
- Check and tune the MV PID loops.
- The control engineer can learn the process dynamics intuitively
  by watching the process step responses.
- Low cost test.

3.3 Test signals for identification

Two aspects of test signals: wave form and frequency content

Signal waveform (time domain) requirements:

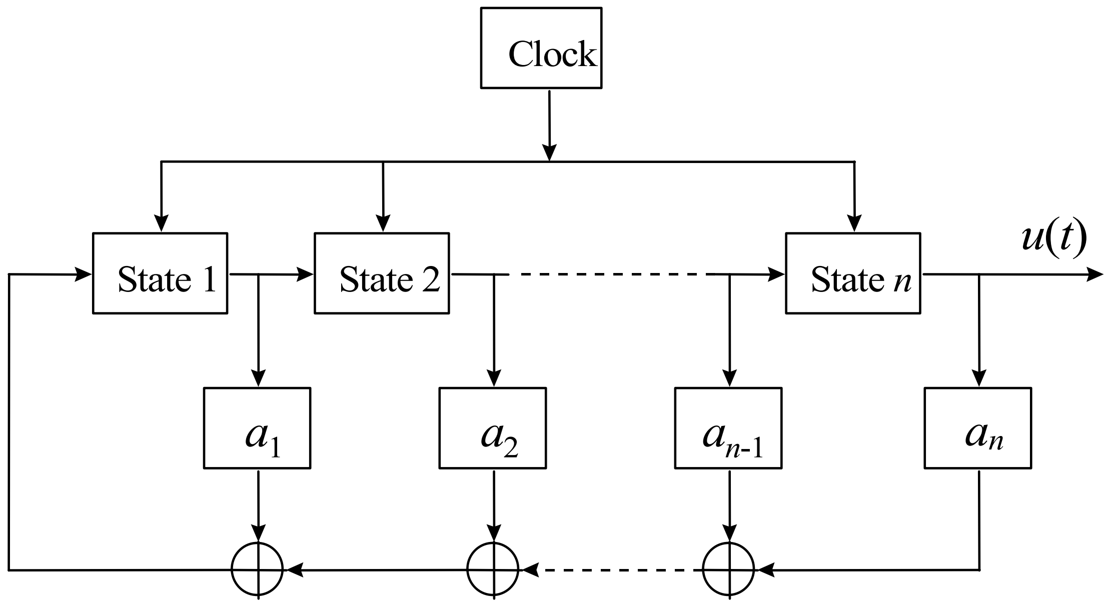
- Not causing upset in normal operation (compare impulse and step).
- Not exciting the process nonlinearity (compare impulse and step).
- Operator friendly (compare impulse and step).

Signal frequency content (power spectrum) design:

- The power spectra of test signals will affect control performance.
- Optimal spectrum design for control purpose.

3.3 Test signals for identification

PRBS (pseudo-random-binary-sequence)



模2加法

x_1	x_2	$x_1 \oplus x_2$
0	0	0
0	1	1
1	0	1
1	1	0

$$x_1 = a_1 x_{i-1} \oplus a_2 x_{i-2} \oplus \cdots \oplus a_n x_{i-n}$$

Maximum period is $M = 2^n - 1$ (M sequence)

3.3 Test signals for identification

PRBS (pseudo-random-binary-sequence)

- **The mean value** $\dfrac{a}{M}$

- **The autocorelation function**

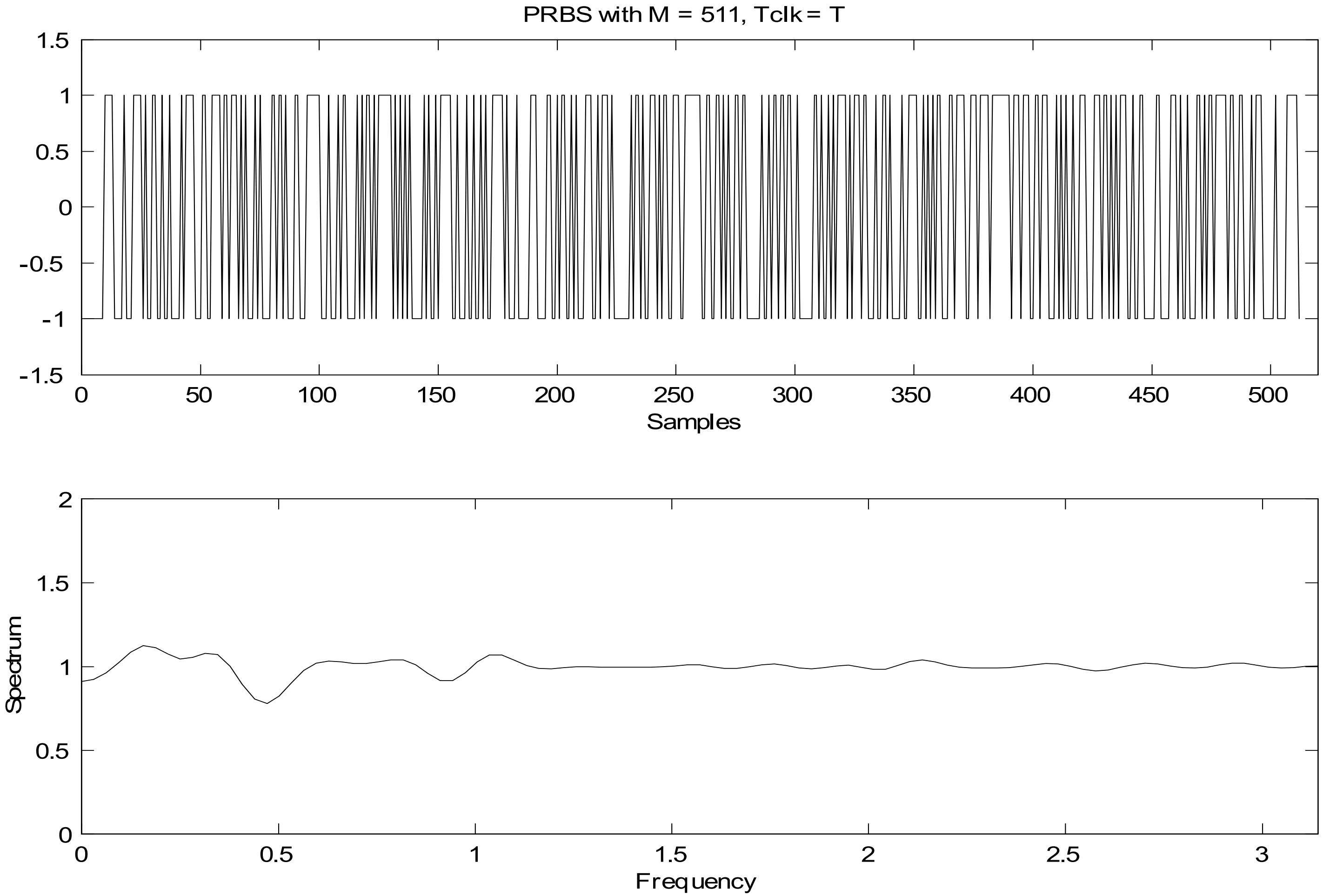
$$R_u(\tau) = \begin{cases} a^2 & \tau = 0, \pm M, \pm 2M, \ldots \\ -\dfrac{a^2}{M} & \text{else} \end{cases}$$

- **The power spectrum**

$$\Phi_u(\omega) = \frac{2\pi a^2}{M} \sum_{k=1}^{M} \delta(\omega - 2\pi k / M), \qquad 0 \le \omega < 2\pi$$

3.3 Test signals for identification

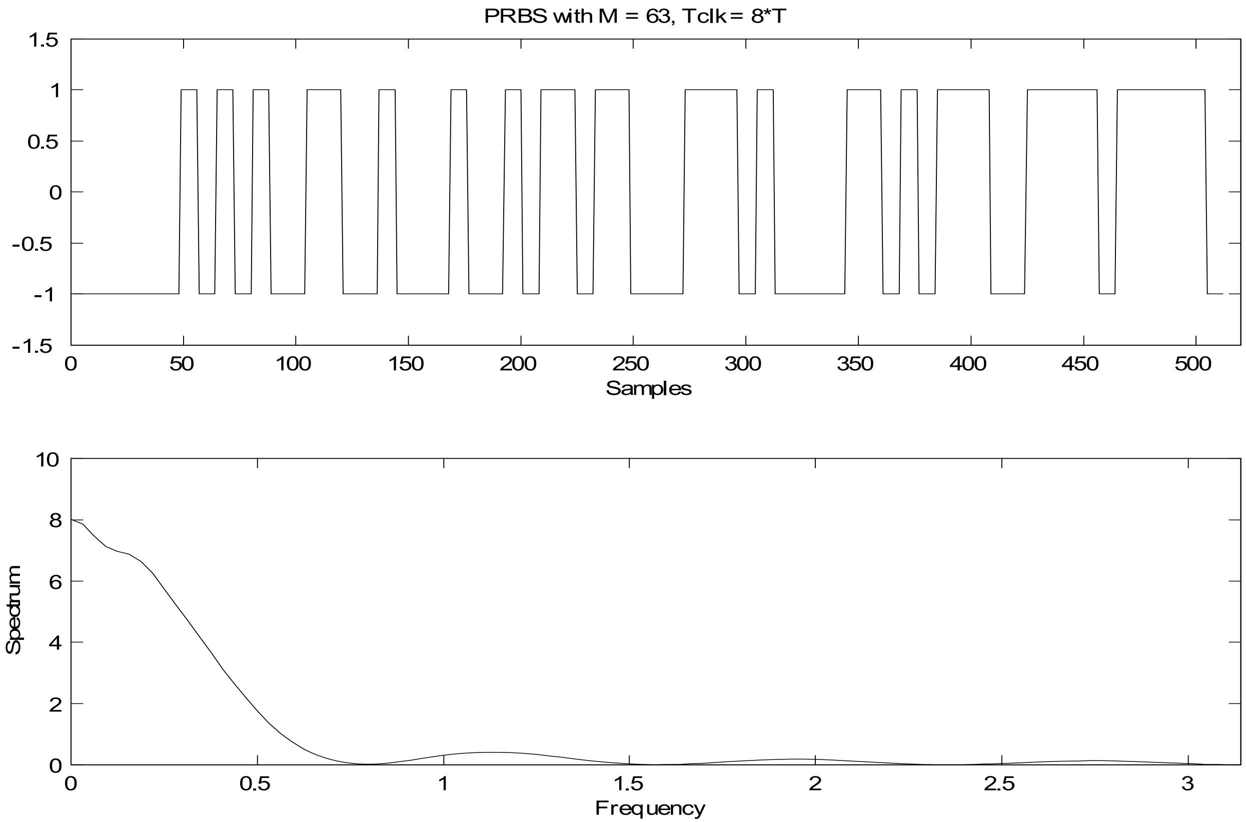
PRBS (pseudo-random-binary-sequence)



3.3 Test signals for identification

Low pass PRBS ($T_{\text{clk}} > T$) power spectrum envelop

$$T_{clk}\left(\frac{\sin(\omega T_{clk}/2)}{\omega T_{clk}/2}\right)^2$$



3.3 Test signals for identification

GBN (generalized binary noise) (Tullenken, 1990)

- Generation rule

$$P[u(t) = -u(t-1)] = p_{\text{sw}}$$

$$P[u(t) = u(t-1)] \quad = 1 - p_{\text{sw}}$$

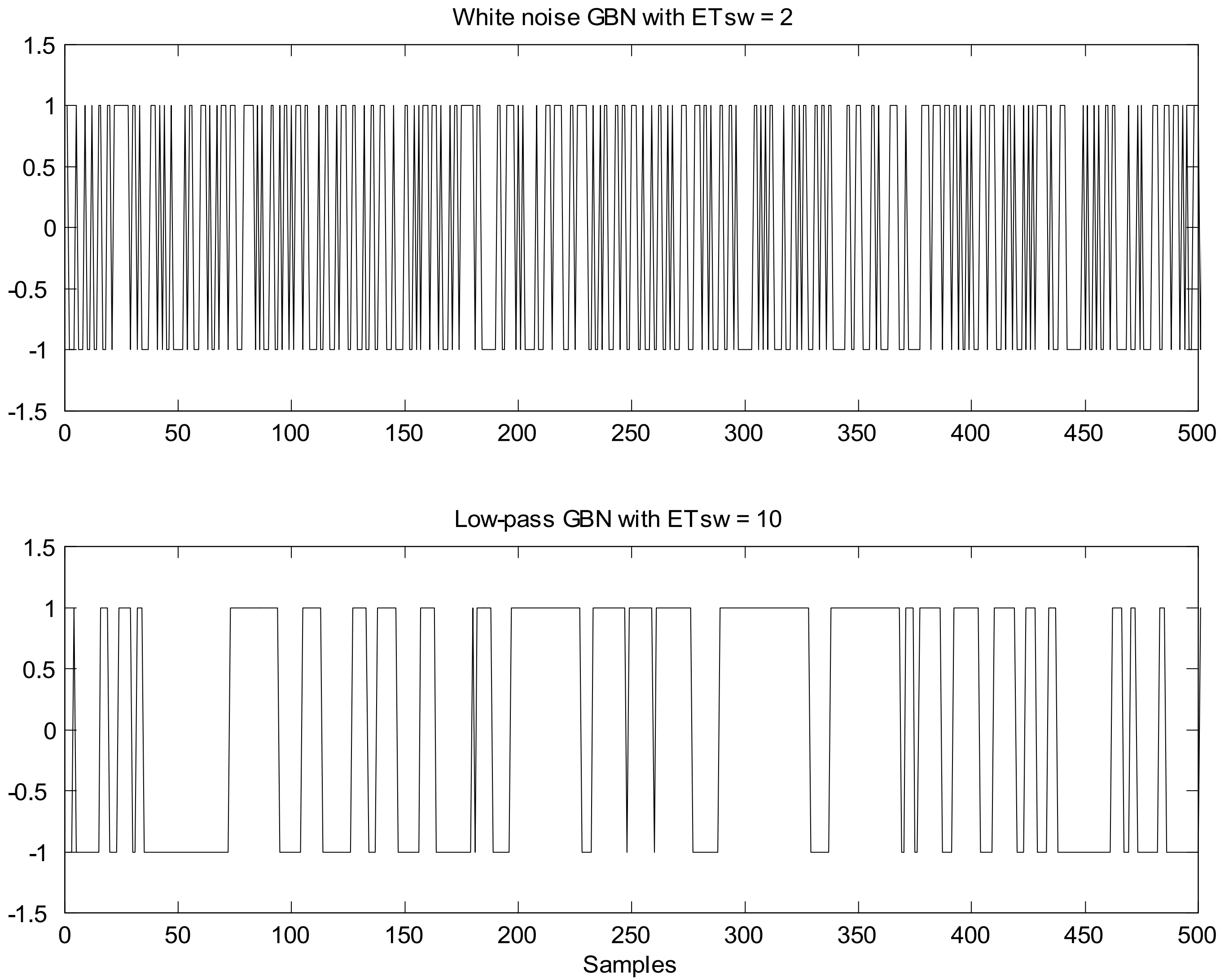
- Mean switching time

$$ET_{\text{sw}} = \frac{T_{\min}}{p_{\text{sw}}}$$

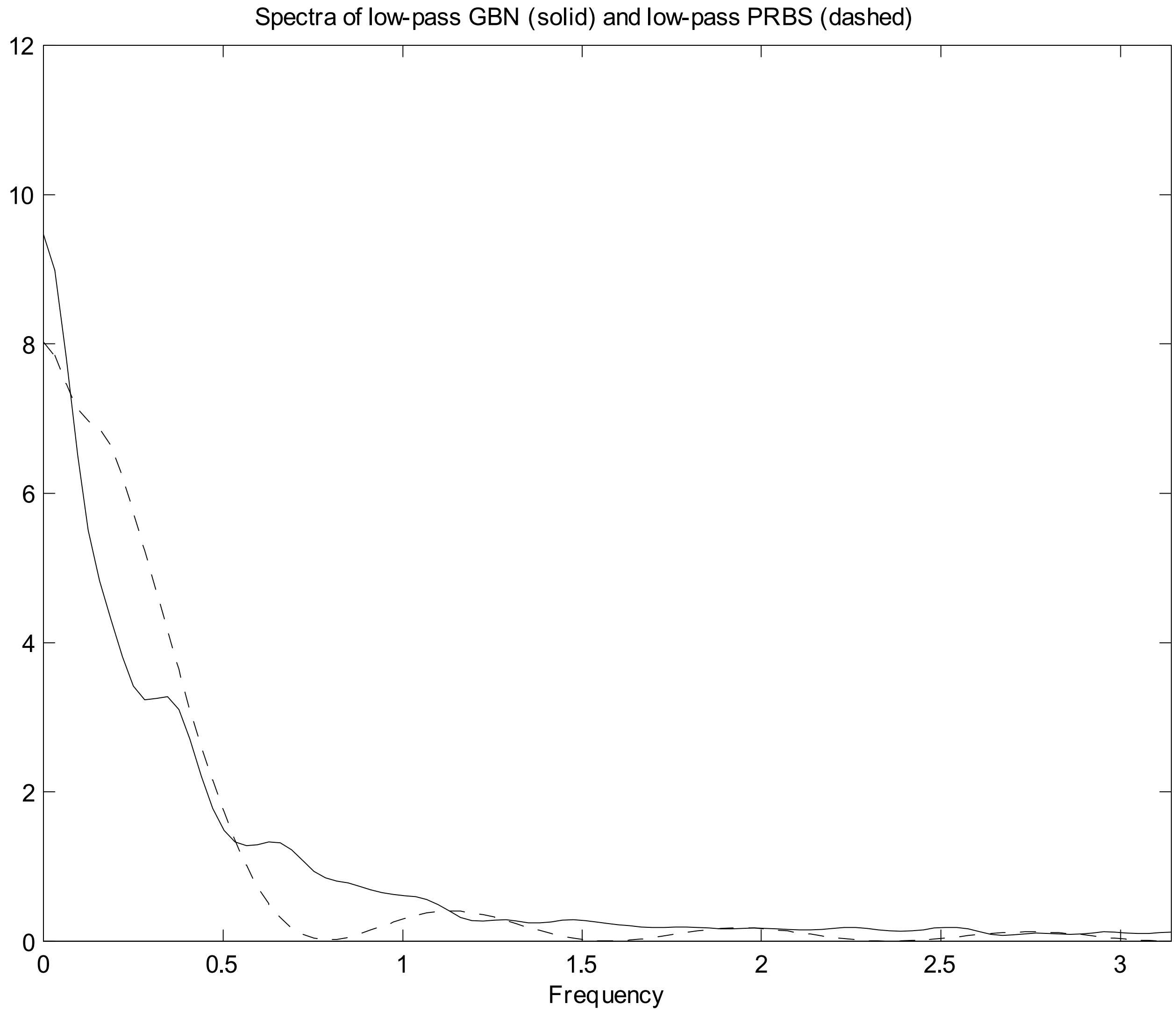
- Power spectrum

$$\Phi_u(\omega) = \frac{(1-q^2)T_{\min}}{1 - 2q\cos T_{\min}\omega + q^2}; \qquad q = 1 - 2p_{\text{sw}}$$

3.3 Test signals for identification



3.3 Test signals for model identification



Chapter 3
Identification Test Design

3.4 Test for model identification, the final test

After the pre-tests, we know:

- The range of normal operation.
- Rough estimate of the process settling time.
- Allowed step sizes of the MVs that will cause CV movements but will not disturb normal process operation.

3.4 Test for model identification, the final test

Test design for MPC control

- Durations of the test:
  Small system or low noise: (6 ~ 10) *(settling time)
  Large system or high noise: (12 ~ 20) *(settling time)
- Signal type: GBN
- Signal step sizes: Determined from pre-tests and discussion with the operator. Start small, can be adjusted during the test.
- Signal mean switching time (for power spectrum):

$$ET_{\mathrm{sw}} = \frac{98\% \text{ settling time}}{3}$$

3.4 Test for model identification, the final test

Test design for MPC control

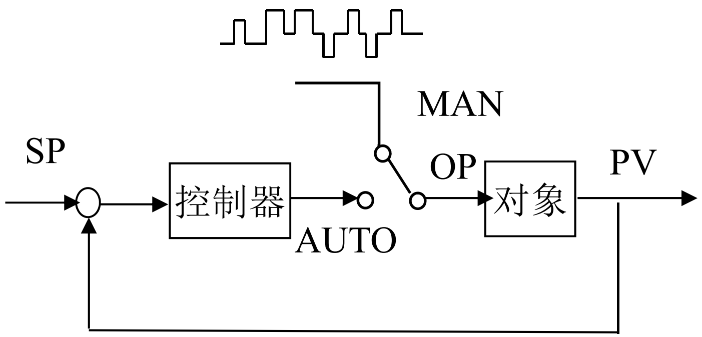
- Number of MVs per test: up 10 MVs in open loop tests; The number MVs per test can be larger if one uses a longer test time.
- Correlation between test signals: Normally independent; for ill-conditioned processes such as high purity distillation columns, strong correlations between some MVs can be helpful.

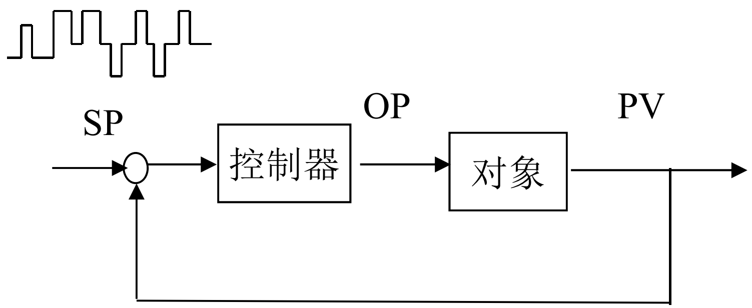
Processes	Mean switching time	Duration of one test
Main fractionator	40 - 60 minutes	40 - 50 hours
Atmospheric tower	40 - 60 minutes	40 - 50 hours
FCC react/regen	15 - 20 minutes	15 - 20 hours
Stabilizer, depropanizer, ...	30 - 50 minutes	30 - 40 hours

3.4 Test for model identification, the final test

Closed-loop tests

- Most industrial application use open loop tests.
- 20 years ago, the industry did not believe the process is identifiable using closed-loop data.
- Test signals can be applied at the setpoints or at the MVs
- Closed-loop test is less disturbing to process operation
- Closed-loop test is easy to carry out.
- Models from closed-loop test data are better for control.





Chapter 4
Identification by the least-squares method

4.1 The principle of least-squares (Gauss, 1809)

Given the model of a system

$$y(t) = x_1(t)\theta_1 + x_2(t)\theta_2 + \cdots + x_n(t)\theta_n$$

and measured data

$$Z^N := x_1(1),\ldots,x_n(1), y(1),\ldots\ldots, x_1(N),\ldots,x_n(N), y(N),$$

Filling the data into the equation

$$y(t) = x_1(t)\theta_1 + x_2(t)\theta_2 + \cdots + x_n(t)\theta_n \quad t = 1, 2, \cdots, N$$

4.1 The principle of least-squares

Introduce the error term (residual)

$$\varepsilon(t) = y(t) - \hat{y}(t) = y(t) - \underbrace{\left[x_1(t)\theta_1 + x_2(t)\theta_2 + \cdots + x_n(t)\theta_n\right]}_{\varphi(t)\theta}$$

Estimate $\hat{\theta}$ by minimizing the loss function (error criterion)

$$V_{LS} = \frac{1}{N}\sum_{t=1}^{N}\varepsilon(t)^2 = \frac{1}{N}\sum_{t=1}^{N}[y(t) - \varphi(t)\theta]^2$$

$$\mathbf{y} = \begin{bmatrix} y(1) \\ y(2) \\ \vdots \\ y(N) \end{bmatrix}, \quad \Phi = \begin{bmatrix} x_1(1) & x_2(1) & \cdots & x_n(1) \\ x_1(t) & x_2(2) & & x_n(2) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(N) & x_2(N) & \cdots & x_n(N) \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix}$$

4.1 The principle of least-squares

In matrix form

$$V_{LS}(\theta) = \frac{1}{N}(\mathbf{y} - \Phi\theta)^T(\mathbf{y} - \Phi\theta)$$

$$= \frac{1}{N}[\mathbf{y}^T\mathbf{y} - \theta^T\Phi^T\mathbf{y} - \mathbf{y}^T\Phi\theta - \theta^T\Phi^T\Phi\theta]$$

Taking the first derivative

$$\frac{\partial V_{LS}(\theta)}{\partial\theta}\Big|_{\theta=\hat{\theta}} = \frac{1}{N}[-2\Phi^T\mathbf{y} + 2\Phi^T\Phi\hat{\theta}] = 0$$

The solution is $\Phi^T\Phi\hat{\theta} = \Phi^T\mathbf{y}$

Normal Equations
正规方程组

$$\hat{\theta} = [\Phi^T\Phi]^{-1}\Phi^T\mathbf{y}$$

4.2 Estimating models of linear processes

Rational transfer functions or ARX models

Given the process model in difference equation

$$y(t)+a_1y(t-1)+\cdots+a_ny(t-n)=b_1u(t-1)+\cdots+b_nu(t-n)$$

Introduce the error term

$$y(t)+a_1y(t-1)+\cdots+a_ny(t-n)=b_1u(t-1)+\cdots+b_nu(t-n)+\varepsilon(t)$$

or

$$A(q)y(t)=B(q)u(t)+\varepsilon(t)$$

The error $\varepsilon(t)$ is called equation error and the model is called ARX

(AutoRegresive with eXoganious input,有源自回归) model

4.2 Estimating models of linear processes

Rational transfer functions or ARX models

Rewrite the equation

$$
\begin{aligned}
y(t) &= -a_1 y(t-1) - \cdots - a_n y(t-n) + b_1 u(t-1) + \cdots + b_n u(t-n) + \varepsilon(t) \\
&= \varphi(t)\theta + \varepsilon(t)
\end{aligned}
$$

where

$$\varphi(t) = [-y(t-1), \cdots - y(t-n) \;\; u(t-1) \cdots u(t-n)]$$

$$\theta = \begin{bmatrix} a_1 \\ \vdots \\ a_n \\ b_1 \\ \vdots \\ b_n \end{bmatrix}$$

4.2 Estimating models of linear processes

Rational transfer functions or ARX models

After the test, we obtain the data set as

$$Z^N = \{y(1),\, u(1),\, \cdots\cdots,\, y(N+n),\, u(N+n)\}$$

Fill in the equation using the data, we have

$$\mathbf{y} = \Phi\theta + \boldsymbol{\varepsilon}$$

where

$$\mathbf{y} = \begin{bmatrix} y(n+1) \\ y(n+2) \\ \vdots \\ y(n+N) \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon(n+1) \\ \varepsilon(n+2) \\ \vdots \\ \varepsilon(n+N) \end{bmatrix}$$

$$\Phi = \begin{bmatrix} \varphi(n+1) \\ \varphi(n+2) \\ \vdots \\ \varphi(n+N) \end{bmatrix} = \left[\begin{array}{ccc|ccc} -y(n) & \cdots & -y(1) & u(n) & \cdots & u(1) \\ -y(n+1) & & \vdots & u(n+1) & & \vdots \\ \vdots & & \vdots & \vdots & & \vdots \\ -y(n+N-1) & \cdots & -y(N) & u(n+N-1) & \cdots & u(N) \end{array}\right]$$

4.2 Estimating models of linear processes

Rational transfer functions or ARX models

The estimator of the model parameters which minimizes

$$V_{LS} = \sum_{t=n+1}^{N+n} \varepsilon(t)^2 = \sum_{t=n+1}^{N+n} [y(t) - \hat{y}(t)]^2 = \sum_{t=n+1}^{N+n} [y(t) - \varphi(t)\theta]^2$$

In matrix form

$$V_{LS}(\theta) = \frac{1}{N}(\mathbf{y} - \Phi\theta)^T(\mathbf{y} - \Phi\theta)$$

$$= \frac{1}{N}[\mathbf{y}^T\mathbf{y} - \theta^T\Phi^T\mathbf{y} - \mathbf{y}^T\Phi\theta - \theta^T\Phi^T\Phi\theta]$$

Taking the first derivative and let it equals to zero

$$\frac{\partial V_{LS}(\theta)}{\partial \theta}\Big|_{\theta=\hat{\theta}} = \frac{1}{N}[-2\Phi^T\mathbf{y} + 2\Phi^T\Phi\hat{\theta}] = 0$$

4.2 Estimating models of linear processes

Rational transfer functions or ARX models

The solution is

$$\Phi^T \Phi \hat{\theta} = \Phi^T \mathbf{y}$$

which given

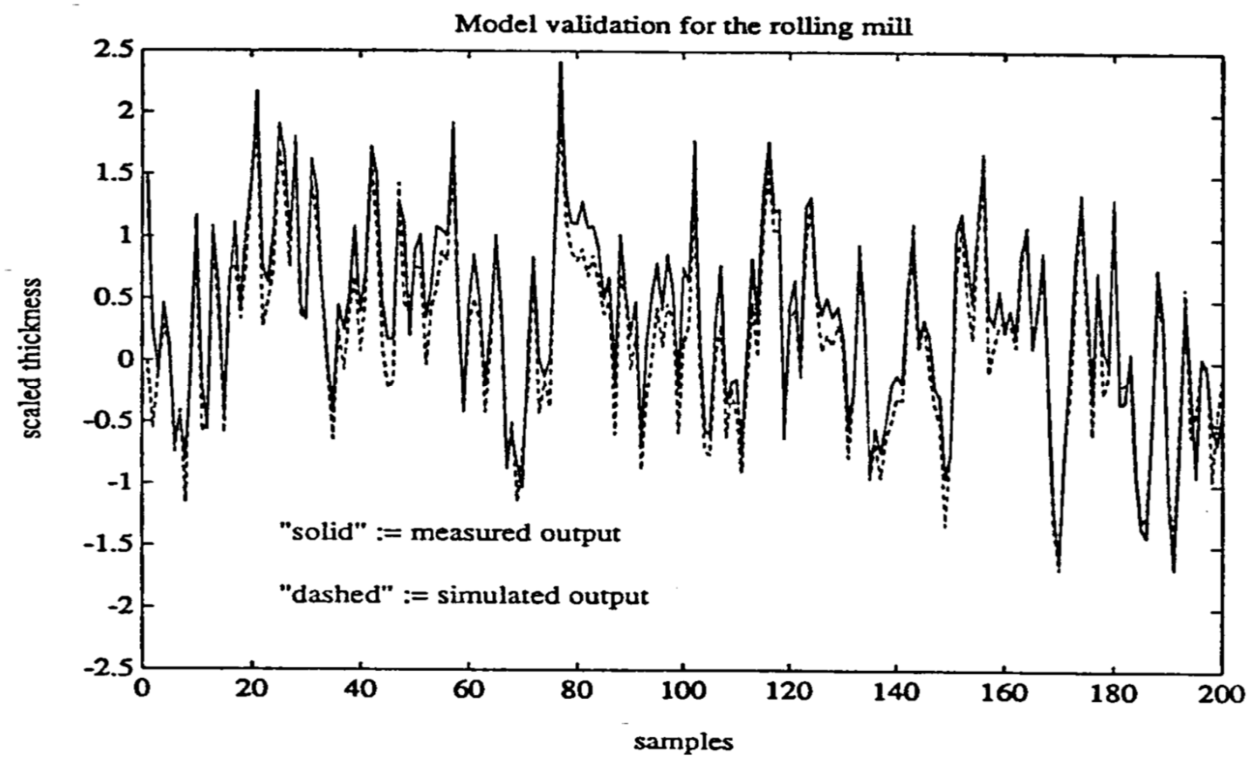
$$\hat{\theta} = [\Phi^T \Phi]^{-1} \Phi^T \mathbf{y} = \left[\sum_{t=n+1}^{N+n} \varphi^T(t)\varphi(t) \right]^{-1} \left[\sum_{t=n+1}^{N+n} \varphi^T(t) y(t) \right]$$

The solution exists if $\Phi^T \Phi = \sum_{t=n+1}^{N+n} \varphi^T(t)\varphi(t)$ is nonsingular

The advantage of ARX model identification by the least-squares method is its numerical simplicity

4.2 Industrial case study

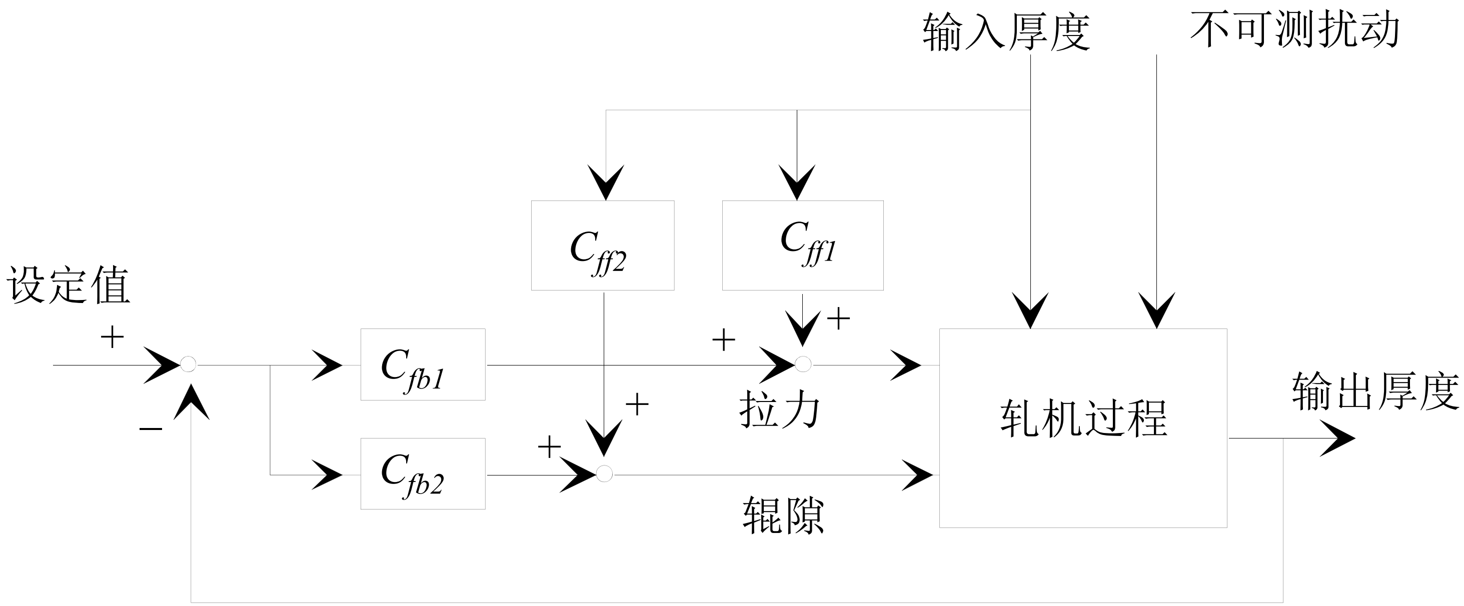
- The identification and control of the rolling mill



error 5%

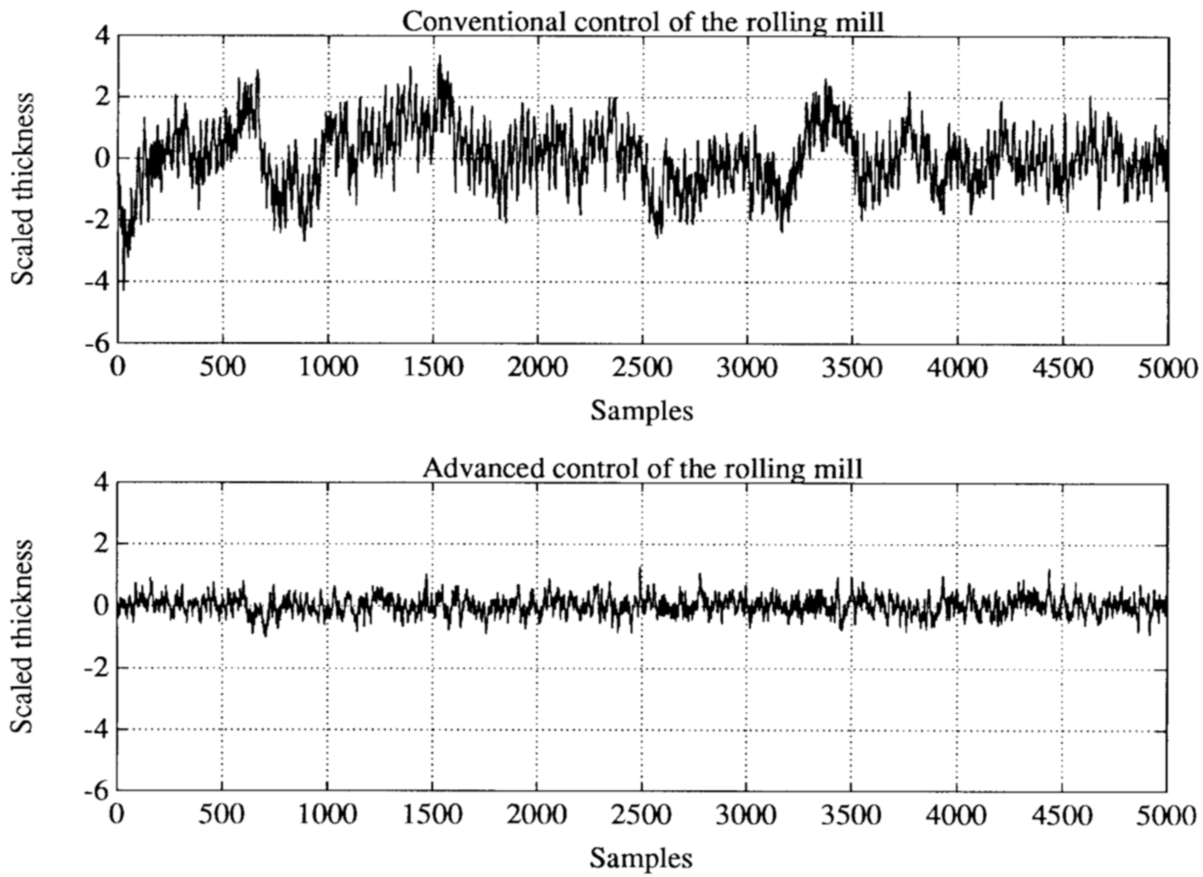
4.2 Industrial case studies

Controller configuration



4.2 Industrial case studies

Control result: over 70% reduction in standard deviation!!



4.2 Estimating models of linear processes

Rational transfer functions or ARX models

Example: Process

$$y(t) = \frac{0.5q^{-1}}{1-0.97q^{-1}}u(t) + v(t)$$

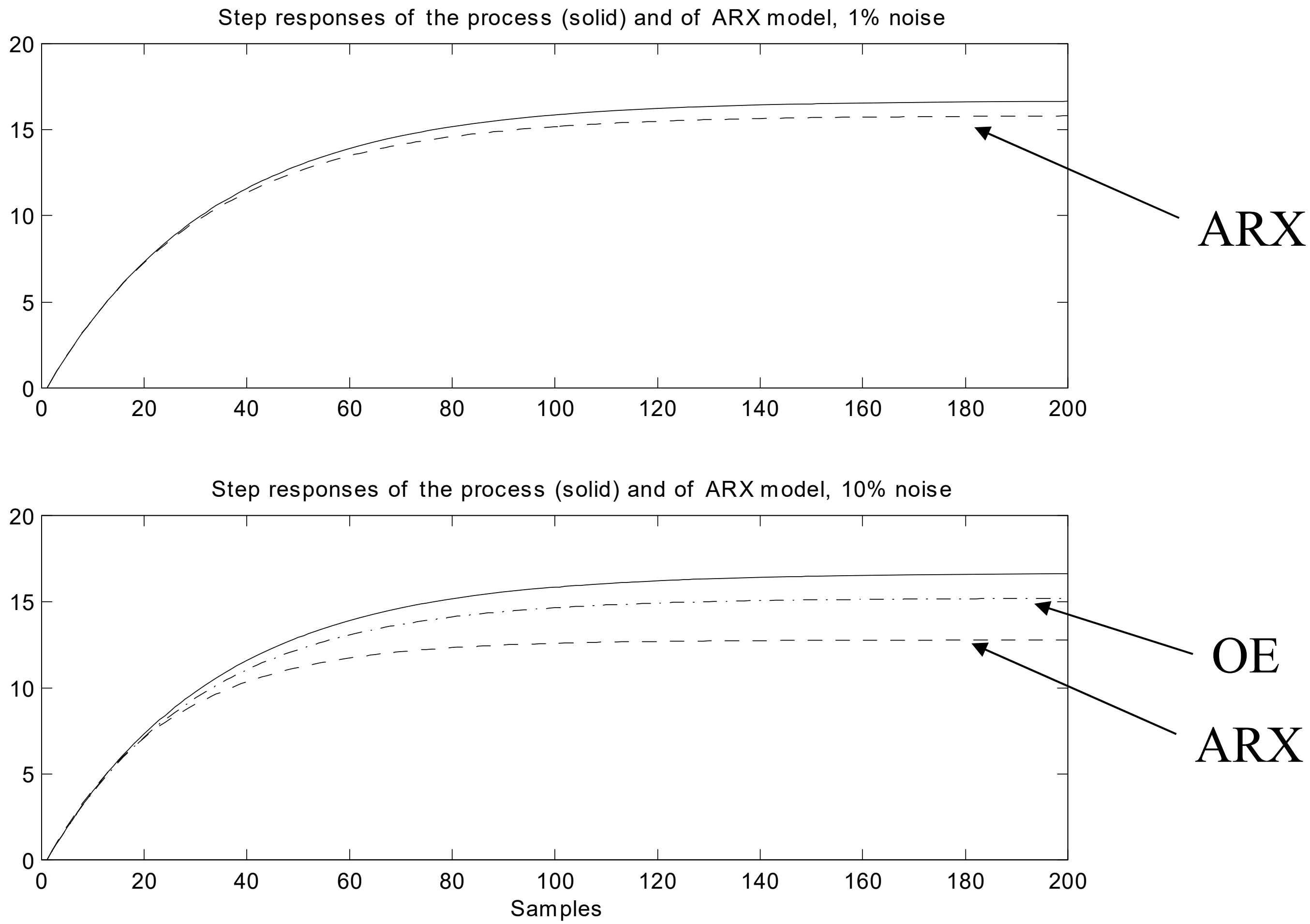
$$v(t) = \frac{1}{1-0.9q^{-1}}e(t)$$

$$\mathrm{var}(v(t)) / \mathrm{var}(y_o(t)) = 1\%,\ 10\%$$

Input: GBN, Tsw = 20 samples,
Number of data N = 1000

4.2 Estimating models of linear processes

Rational transfer functions or ARX models



4.4 Properties of the least-squares estimator

Because the noise term $e(t)$ is assumed a realization of a stochastic process $\{e(t)\}$, the estimate $\hat{\theta}$ is a random variable.

Therefore, the analysis of identification methods is done in a probability framework.

4.4 Properties of the least-squares estimator

Desirable (good) properties of an estimator (ID method)

1) Unbiased estimator (model): if for each sample number N

$$E\hat{\theta} = \theta^o$$

2) Consistent estimator:

$$\text{if } N \to \infty \text{ then } \hat{\theta} \to \theta^o \text{ with probability 1}$$

3) Efficient or minimum variance estimator: if for all unbiased estimators $\hat{\theta}^*$

$$\text{cov}(\hat{\theta}) \le \text{cov}(\hat{\theta}^*), \text{ or, } \det[\text{cov}(\hat{\theta})] - \det[\text{cov}(\hat{\theta}^*)] \le 0$$

where

$$\text{cov}(\hat{\theta}) = E\{[\hat{\theta} - E(\hat{\theta})][\hat{\theta} - E(\hat{\theta})]^T\}$$

4.4 Properties of the least-squares estimator

Assume that the data are generated by the true process:

$$y(t) = \varphi(t)\theta^o + e(t)$$

where

$$\varphi(t) = [-y(t-1),\cdots - y(t-n) \;\; u(t-1)\cdots u(t-n)] \quad \theta^o = \begin{bmatrix} \theta_1^o \\ \theta_2^o \\ \vdots \\ \theta_n^o \end{bmatrix}$$

Assumptions:

A1: The error term $\{e(t)\}$ is a stationary stochastic process with zero mean ($Ee(t) = 0$)

A2: $e(t)$ and data vector $\varphi(t)$ are uncorrelated ($E\varphi^T(t)e(t) = 0$)

4.4 Properties of the least-squares estimator

Substituting $y(t) = \varphi(t)\theta^o + e(t)$ into $\hat{\theta} = [\Phi^T \Phi]^{-1} \Phi^T \mathbf{y}$

yields
$$\hat{\theta} = \theta^o + [\Phi^T \Phi]^{-1} \Phi^T \mathbf{e}$$

Then, **under A1 and A2**

$$E\hat{\theta} = E\theta^o + E[(\Phi^T \Phi)^{-1} \Phi^T \mathbf{e}]$$

$$\Downarrow$$

$$E\hat{\theta} = E\theta^o + E[(\Phi^T \Phi)^{-1} \Phi^T]E(\mathbf{e}) = \theta^o$$

Hence the *original* least-squares estimator is **unbiased.**

Properties of the ARX model

Assume that the true process is given as

$$y(t) = \frac{B^o(q)}{A^o(q)} u(t) + v(t)$$

输出扰动，零均值的平稳随机过程

Rewrite this relation in the ARX model

$$A^o(q) y(t) = B^o(q) u(t) + A^o(q) v(t)$$

$$\Rightarrow A^o(q) y(t) = B^o(q) u(t) + e(t), \quad e(t) = A^o(q) v(t)$$

$$\Rightarrow y(t) = \varphi(t) \theta^o + e(t)$$

$$\varphi(t) = [-y(t-1), \cdots - y(t-n) \;\; u(t-1) \cdots u(t-n)]$$

Now $e(t)$ and $y(t\text{-}1)$, …, $y(t\text{-}n)$ are correlated because output y(t) contains the disturbance v(t) and e(t) is correlated with v(t) via

$$e(t) = v(t) + a_1 v(t-1) + \ldots + a_n v(t-n)$$

Conclusion: **The ARX model is biased because A2 does not hold**.

Properties of the ARX model

When the true process **has** an ARX structure:

$$y(t) = \frac{B^o(q)}{A^o(q)} u(t) + \frac{1}{A^o(q)} e(t)$$

$$A^o(q) y(t) = B^o(q) u(t) + e(t)$$

where $e(t)$ is zero mean white noise with variance σ^2

Now $e(t)$ and $y(t\text{-}1), \ldots, y(t\text{-}n)$ are NOT correlated

Then A2 holds and the ARX model is unbiased and minimum variance